Network Traffic Analysis with Wireshark

1. Objective

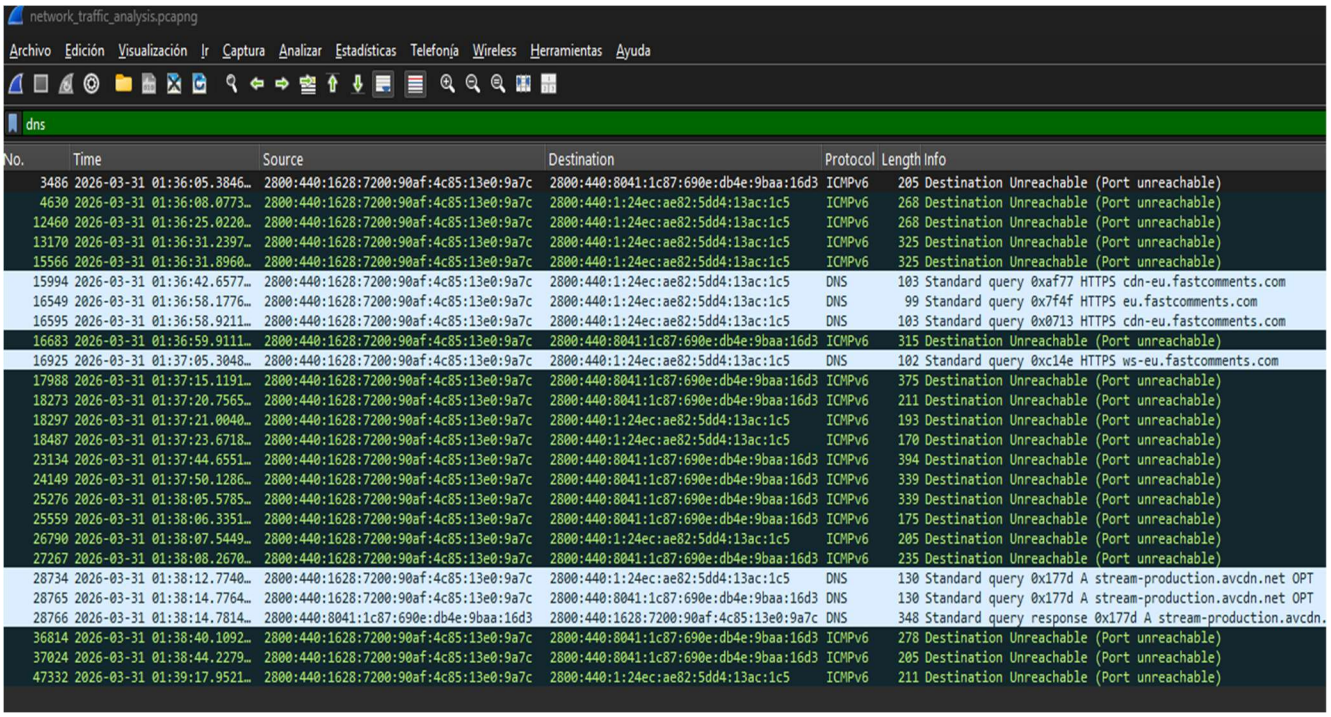
Analyze network traffic using Wireshark to identify protocols, DNS queries, TCP connections, and secure communications.

2. Environment

Tool: Wireshark
Network: Local network
Capture file: network_traffic_analysis.pcapng

3. DNS Analysis

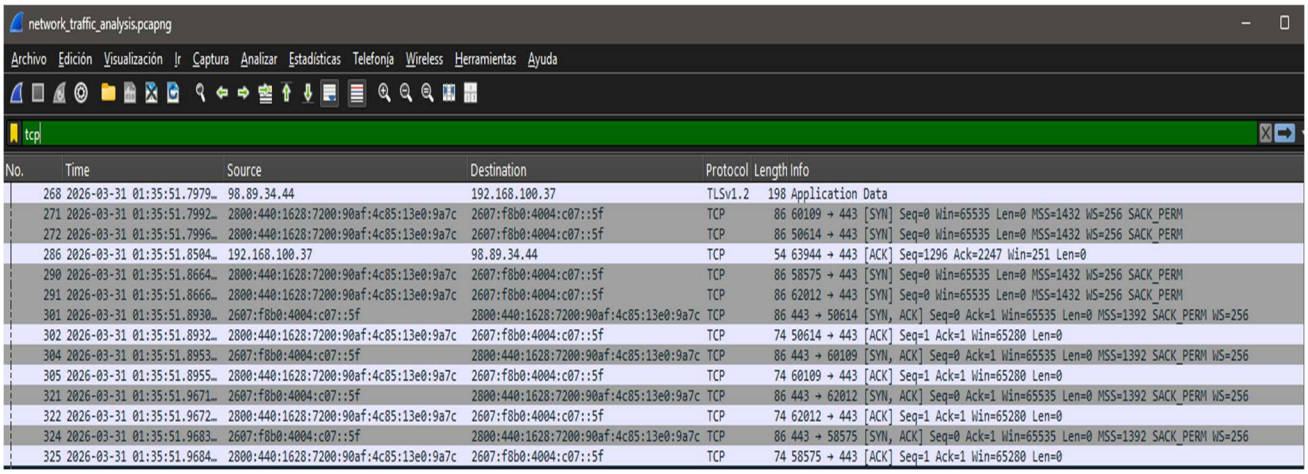
DNS traffic was analyzed to identify domain resolution requests. Queries such as cdn-eu.fastcomments.com and stream-production.avcdn.net were observed.



No.	Time	Source	Destination	Protocol	Length	Info
3486	2026-03-31 01:36:05.3846…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	205	Destination Unreachable (Port unreachable)
4630	2026-03-31 01:36:08.0773…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	ICMPv6	268	Destination Unreachable (Port unreachable)
12460	2026-03-31 01:36:25.0220…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	ICMPv6	268	Destination Unreachable (Port unreachable)
13170	2026-03-31 01:36:31.2397…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	ICMPv6	325	Destination Unreachable (Port unreachable)
15566	2026-03-31 01:36:31.8960…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	ICMPv6	325	Destination Unreachable (Port unreachable)
15994	2026-03-31 01:36:42.6577…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	DNS	103	Standard query 0xaf77 HTTPS cdn-eu.fastcomments.com
16549	2026-03-31 01:36:58.1776…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	DNS	99	Standard query 0x7f4f HTTPS eu.fastcomments.com
16595	2026-03-31 01:36:58.9211…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	DNS	103	Standard query 0x0713 HTTPS cdn-eu.fastcomments.com
16683	2026-03-31 01:36:59.9111…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	315	Destination Unreachable (Port unreachable)
16925	2026-03-31 01:37:05.3048…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	DNS	102	Standard query 0xc14e HTTPS ws-eu.fastcomments.com
17988	2026-03-31 01:37:15.1191…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	375	Destination Unreachable (Port unreachable)
18273	2026-03-31 01:37:20.7565…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	211	Destination Unreachable (Port unreachable)
18297	2026-03-31 01:37:21.0040…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	ICMPv6	193	Destination Unreachable (Port unreachable)
18487	2026-03-31 01:37:23.6718…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	ICMPv6	170	Destination Unreachable (Port unreachable)
23134	2026-03-31 01:37:44.6551…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	394	Destination Unreachable (Port unreachable)
24149	2026-03-31 01:37:50.1286…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	339	Destination Unreachable (Port unreachable)
25276	2026-03-31 01:38:05.5785…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	339	Destination Unreachable (Port unreachable)
25559	2026-03-31 01:38:06.3351…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	175	Destination Unreachable (Port unreachable)
26790	2026-03-31 01:38:07.5449…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	ICMPv6	205	Destination Unreachable (Port unreachable)
27267	2026-03-31 01:38:08.2670…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	235	Destination Unreachable (Port unreachable)
28734	2026-03-31 01:38:12.7740…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	DNS	130	Standard query 0x177d A stream-production.avcdn.net OPT
28765	2026-03-31 01:38:14.7764…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	DNS	130	Standard query 0x177d A stream-production.avcdn.net OPT
28766	2026-03-31 01:38:14.7814…	2800:440:8041:1c87:690e:db4e:9baa:16d3	2800:440:1628:7200:90af:4c85:13e0:9a7c	DNS	348	Standard query response 0x177d A stream-production.avcdn.
36814	2026-03-31 01:38:40.1092…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	278	Destination Unreachable (Port unreachable)
37024	2026-03-31 01:38:44.2279…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:8041:1c87:690e:db4e:9baa:16d3	ICMPv6	205	Destination Unreachable (Port unreachable)
47332	2026-03-31 01:39:17.9521…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2800:440:1:24ec:ae82:5dd4:13ac:1c5	ICMPv6	211	Destination Unreachable (Port unreachable)

4. TCP Analysis

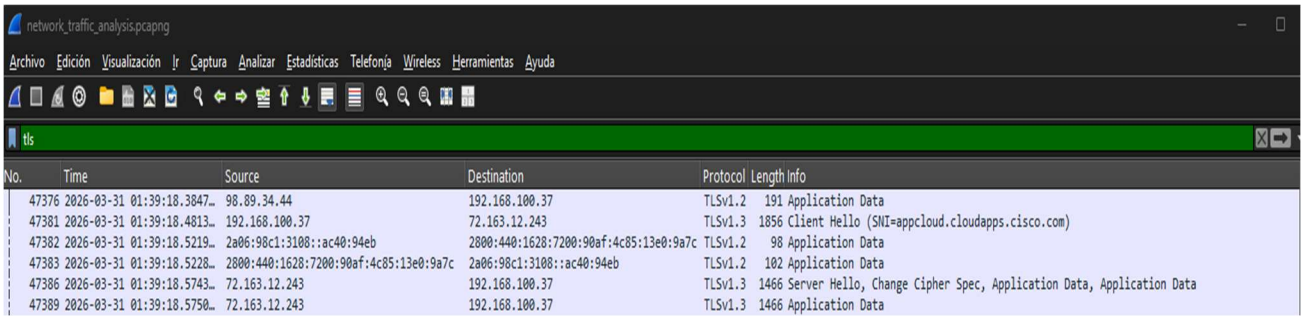
TCP analysis revealed the three-way handshake (SYN, SYN-ACK, ACK), confirming successful connections. Most traffic was directed to port 443 (HTTPS).



No.	Time	Source	Destination	Protocol	Length	Info
268	2026-03-31 01:35:51.7979…	98.89.34.44	192.168.100.37	TLSv1.2	198	Application Data
271	2026-03-31 01:35:51.7992…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2607:f8b0:4004:c07::5f	TCP	86	60109 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
272	2026-03-31 01:35:51.7996…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2607:f8b0:4004:c07::5f	TCP	86	50614 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
286	2026-03-31 01:35:51.8504…	192.168.100.37	98.89.34.44	TCP	54	63944 → 443 [ACK] Seq=1296 Ack=2247 Win=251 Len=0
290	2026-03-31 01:35:51.8664…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2607:f8b0:4004:c07::5f	TCP	86	58575 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
291	2026-03-31 01:35:51.8666…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2607:f8b0:4004:c07::5f	TCP	86	62012 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
301	2026-03-31 01:35:51.8930…	2607:f8b0:4004:c07::5f	2800:440:1628:7200:90af:4c85:13e0:9a7c	TCP	86	443 → 50614 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1392 SACK_PERM WS=256
302	2026-03-31 01:35:51.8932…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2607:f8b0:4004:c07::5f	TCP	74	50614 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
304	2026-03-31 01:35:51.8953…	2607:f8b0:4004:c07::5f	2800:440:1628:7200:90af:4c85:13e0:9a7c	TCP	86	443 → 60109 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1392 SACK_PERM WS=256
305	2026-03-31 01:35:51.8955…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2607:f8b0:4004:c07::5f	TCP	74	60109 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
321	2026-03-31 01:35:51.9671…	2607:f8b0:4004:c07::5f	2800:440:1628:7200:90af:4c85:13e0:9a7c	TCP	86	443 → 62012 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1392 SACK_PERM WS=256
322	2026-03-31 01:35:51.9672…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2607:f8b0:4004:c07::5f	TCP	74	62012 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
324	2026-03-31 01:35:51.9683…	2607:f8b0:4004:c07::5f	2800:440:1628:7200:90af:4c85:13e0:9a7c	TCP	86	443 → 58575 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1392 SACK_PERM WS=256
325	2026-03-31 01:35:51.9684…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2607:f8b0:4004:c07::5f	TCP	74	58575 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0

5. TLS Analysis

TLS traffic showed encrypted communication including Client Hello and Server Hello messages. SNI revealed domains such as appcloud.cloudapps.cisco.com.



No.	Time	Source	Destination	Protocol	Length	Info
47376	2026-03-31 01:39:18.3847…	98.89.34.44	192.168.100.37	TLSv1.2	191	Application Data
47381	2026-03-31 01:39:18.4813…	192.168.100.37	72.163.12.243	TLSv1.3	1856	Client Hello (SNI=appcloud.cloudapps.cisco.com)
47382	2026-03-31 01:39:18.5219…	2a06:98c1:3108::ac40:94eb	2800:440:1628:7200:90af:4c85:13e0:9a7c	TLSv1.2	98	Application Data
47383	2026-03-31 01:39:18.5228…	2800:440:1628:7200:90af:4c85:13e0:9a7c	2a06:98c1:3108::ac40:94eb	TLSv1.2	102	Application Data
47386	2026-03-31 01:39:18.5743…	72.163.12.243	192.168.100.37	TLSv1.3	1466	Server Hello, Change Cipher Spec, Application Data, Application Data
47389	2026-03-31 01:39:18.5750…	72.163.12.243	192.168.100.37	TLSv1.3	1466	Application Data

6. Conversations Analysis

Multiple TCP conversations were identified between the local host and external servers over HTTPS, indicating normal browsing behavior.

Ethernet · 1 | IPv4 · 40 | IPv6 · 66 | TCP · 191 | UDP · 20

Dirección A	Puerto A	Dirección B	Puerto B	Paquetes	Bytes	Stream ID	Paquetes totales	Porcentaje filtrado	Packets A → B	Bytes A → B
192.168.100.37	53633	16.145.251.96	443	21	15 kB	109	44	47.73%	9	11 kB
192.168.100.37	54422	18.160.10.110	443	9	7 kB	200	24	37.50%	4	3 kB
192.168.100.37	64863	18.210.210.118	443	1	78 bytes	95	4	25.00%	1	78 bytes
192.168.100.37	57273	20.42.73.30	443	20	73 kB	79	61	32.79%	10	69 kB
192.168.100.37	62445	20.44.239.154	443	10	3 kB	121	22	45.45%	5	2 kB
192.168.100.37	62341	20.75.60.91	443	11	7 kB	183	31	35.48%	5	4 kB
192.168.100.37	59860	20.190.190.193	443	6	5 kB	85	16	37.50%	3	2 kB
192.168.100.37	62955	20.190.190.193	443	6	8 kB	71	23	26.09%	3	7 kB
192.168.100.37	53829	23.12.65.77	443	15	7 kB	143	35	42.86%	5	4 kB
192.168.100.37	49225	32.194.100.119	443	6	498 bytes	39	12	50.00%	6	498 bytes
192.168.100.37	62493	34.111.24.1	443	1	127 bytes	63	12	8.33%	0	0 bytes
192.168.100.37	57433	34.199.156.218	443	5	415 bytes	22	12	41.67%	5	415 bytes
192.168.100.37	65505	34.218.218.254	443	35	8 kB	41	56	62.50%	14	4 kB
192.168.100.37	64656	35.153.78.52	443	9	6 kB	96	22	40.91%	4	3 kB
192.168.100.37	50635	35.199.147.118	443	8	6 kB	118	24	33.33%	3	3 kB
192.168.100.37	53756	35.199.147.118	443	7	9 kB	217	17	41.18%	3	5 kB
192.168.100.37	55785	35.199.147.118	443	7	9 kB	218	17	41.18%	3	5 kB
192.168.100.37	55998	35.199.147.118	443	8	6 kB	113	22	36.36%	3	2 kB
192.168.100.37	56792	35.199.147.118	443	7	6 kB	213	14	50.00%	3	2 kB
192.168.100.37	58057	35.199.147.118	443	7	6 kB	214	14	50.00%	3	2 kB
192.168.100.37	59357	35.199.147.118	443	8	6 kB	114	22	36.36%	3	2 kB
192.168.100.37	62443	35.199.147.118	443	8	6 kB	119	24	33.33%	3	3 kB
192.168.100.37	54837	52.87.15.58	443	10	10 kB	108	25	40.00%	3	5 kB
192.168.100.37	54960	52.168.117.168	443	15	10 kB	146	31	48.39%	8	8 kB
192.168.100.37	59729	54.245.47.218	443	11	3 kB	44	30	36.67%	3	2 kB
192.168.100.37	54760	57.145.2.145	443	50	19 kB	61	76	65.79%	27	12 kB
192.168.100.37	49271	63.140.38.61	443	4	4 kB	203	14	28.57%	2	2 kB
192.168.100.37	55890	63.140.38.61	443	6	4 kB	174	23	26.09%	3	2 kB
192.168.100.37	56835	63.140.38.61	443	5	4 kB	27	22	22.73%	2	2 kB
192.168.100.37	59896	63.140.38.61	443	6	4 kB	100	22	27.27%	3	2 kB
192.168.100.37	49944	63.140.38.107	443	6	4 kB	175	23	26.09%	3	2 kB
192.168.100.37	63569	63.140.38.107	443	6	4 kB	102	23	26.09%	3	2 kB

7. Conclusion

The analysis demonstrates typical secure network communication, including DNS resolution, TCP connection establishment, and encrypted TLS traffic.